

CONTROL by Control AI

Comprehensive Report of Functions, Features, and Capabilities

Generated: March 18, 2026

EXECUTIVE SUMMARY

Control is a revolutionary AI-powered desktop automation platform developed by Control AI. It represents a paradigm shift in human-computer interaction by enabling users to control their entire computer through natural language voice or text commands. The system comprises two integrated components: a desktop application built on Electron for local automation, and a web platform for cloud-based virtual machine management with remote desktop capabilities.

The platform's core innovation lies in its ability to 'see' and 'act' on any desktop application using computer vision and sophisticated AI reasoning, eliminating the need for application-specific APIs or plugins. This universal approach allows Control to operate any software from complex professional tools like AutoCAD and Blender to simple system utilities.

SYSTEM ARCHITECTURE

Desktop Application (Electron-based)

The desktop application serves as the primary interface for local computer automation. Built on the Electron framework, it provides cross-platform compatibility across macOS, Windows, and Linux. The architecture follows a modular design pattern with distinct managers handling specific functionalities:

- **ComputerUseAgent (Main Controller):** The central orchestrator that coordinates all subsystems including window management, hotkey handling, security protocols, backend communication, and wakeword detection.
- **WindowManager:** Manages multiple application windows including the main chat interface, settings panel, workflow editor, and entry overlay.
- **HotkeyManager:** Handles global keyboard shortcuts for quick activation of Control features from anywhere in the operating system.
- **SecurityManager:** Implements security protocols including sensitive action approvals, terminal command permissions, and user consent mechanisms.
- **BackendManager:** Routes requests between the ASK mode (conversational) and ACT mode (automation) backends based on user intent.

- **WakewordManager:** Enables hands-free activation through the 'Hey Control' wake word using Picovoice Porcupine detection engine.
- **EdgeTTSManager:** Provides text-to-speech capabilities with both online (Edge TTS) and offline fallback options.
- **VoskServerManager:** Manages local speech-to-text transcription for voice commands.
- **WorkflowManager:** Handles automation workflow creation, storage, and execution with keyword and time-based triggers.
- **RemoteDesktopManager:** Enables connection to cloud-hosted virtual machines through the web platform.

Web Application (Next.js + FastAPI)

The web platform extends Control's capabilities to cloud infrastructure, enabling users to provision and manage virtual machines with AI automation capabilities. The architecture consists of:

- **Frontend (Next.js 14):** React-based dashboard with TypeScript for type safety, Tailwind CSS for styling, Framer Motion for animations, and Zustand for state management.
- **Backend (FastAPI):** Python-based REST API handling VM lifecycle management, user authentication via Supabase, payment processing through Flutterwave, and device pairing.
- **Remote Desktop Viewer:** VNC/noVNC integration for real-time remote desktop streaming within the browser.
- **AI Agent (Python):** WebSocket-based automation agent running inside VMs for cloud-based task execution.

CORE FUNCTIONS AND CAPABILITIES

1. Visual AI and Screen Understanding

Control's visual AI system represents a breakthrough in desktop automation. Unlike traditional automation tools that rely on brittle coordinate recording or application-specific APIs, Control uses computer vision to interpret screen pixels in real-time:

- **Screenshot Analysis:** Captures high-resolution screenshots with cursor position marking for precise action targeting.
- **UI Element Detection:** Identifies buttons, sliders, menus, text fields, and other interactive elements through pixel analysis.
- **Normalized Coordinate System:** Maps screen positions to a 1000x1000 grid, enabling consistent targeting across different screen resolutions.
- **Multi-Monitor Support:** Extends visual understanding across multiple display configurations.
- **Dynamic Interface Adaptation:** Responds to UI changes in real-time without requiring pre-programmed templates.

2. Mouse and Keyboard Automation

The automation engine provides comprehensive control over input devices through `@computer-use/nut-js`:

- **Mouse Actions:** click, double_click, right_click, mouse_move, drag, scroll with pixel-perfect precision.
- **Keyboard Actions:** type (with configurable speed), key_press for shortcuts and special keys.
- **Terminal Execution:** Run shell commands with output capture and error handling.
- **Clipboard Operations:** Read from and write to system clipboard for data transfer.
- **Anti-Detection Patterns:** Human-like movement curves and randomized timing for applications sensitive to automation.

3. Voice Control System

Control features a sophisticated voice control pipeline enabling hands-free operation:

- **Wake Word Detection:** 'Hey Control' activation using Picovoice Porcupine engine with high accuracy and low false-positive rate.
- **Speech-to-Text:** Vosk server integration for local, privacy-preserving transcription.
- **Text-to-Speech:** Edge TTS with multiple voice options (Jenny, Aria, Guy, etc.) and configurable rate/volume.
- **Offline Fallback:** Automatic switching to offline TTS when network connectivity is unavailable.
- **Push-to-Talk:** Alternative activation method via configurable keyboard shortcut.

4. Multi-Provider AI Integration

Control supports multiple AI providers for maximum flexibility and redundancy:

- **Google Gemini** (Primary): High-performance multimodal model with excellent visual understanding.
- **Anthropic Claude**: Advanced reasoning capabilities for complex task planning.
- **OpenAI GPT-4**: Versatile language model with strong instruction following.
- **DeepSeek**: Cost-effective alternative for routine tasks.
- **Ollama**: Local model support for privacy-sensitive environments.
- **OpenRouter**: Unified API access to multiple models.
- **Groq**: High-speed inference for real-time applications.

5. Browser Automation

Built-in browser automation capabilities extend Control's reach to web applications:

- **Browser Navigation:** Open, navigate, and manage browser sessions.
- **JavaScript Execution:** Run custom JavaScript for advanced web automation.
- **Web Scraping:** Extract data from web pages with intelligent parsing.
- **Form Interaction:** Fill forms, click buttons, and navigate complex web workflows.
- **Session Persistence:** Maintain login states and cookies across sessions.

6. Workflow Automation

The workflow system enables users to create and save automated sequences:

- **Keyword Triggers:** Execute workflows when specific phrases are detected.
- **Time-Based Triggers:** Schedule workflows at specific times or intervals.
- **Import/Export:** Share workflows with team members or across devices.
- **Action Libraries:** Pre-built workflow templates for common tasks.
- **Conditional Logic:** Branch workflows based on conditions and results.

7. Cloud VM Management

The web platform provides enterprise-grade virtual machine management:

- **VM Provisioning:** Create virtual machines with customizable specifications.
- **Remote Desktop:** VNC/noVNC streaming for browser-based remote access.
- **AI Agent Deployment:** Install automation agents on VMs for cloud-based execution.
- **Session Management:** Track and manage active automation sessions.
- **Resource Monitoring:** Real-time status and health indicators for VMs.

8. Security and Privacy

Control implements robust security measures to protect user data and system integrity:

- **Sensitive Action Approval:** User confirmation required for potentially destructive operations.
- **Terminal Permission System:** Explicit approval for shell command execution.
- **Local-First Architecture:** Minimizes data transmission to cloud services.
- **Supabase Authentication:** Enterprise-grade user authentication and session management.
- **Encrypted Communications:** All API calls and data transfers use encryption.

TECHNICAL SPECIFICATIONS

Desktop Application Stack

- Framework: Electron (Cross-platform desktop)
- Language: JavaScript (ES6+)
- UI: HTML/CSS with custom theming
- Automation: @computer-use/nut-js
- AI Integration: @google/generative-ai, @anthropic-ai/sdk, openai
- Voice Detection: @picovoice/porcupine-node, @picovoice/pvrecorder-node
- TTS: edge-tts (Python), say (fallback)
- STT: Vosk server integration
- Database: Supabase (PostgreSQL)
- Build: electron-builder for macOS, Windows, Linux

Web Application Stack

- Frontend: Next.js 14 with TypeScript
- Styling: Tailwind CSS
- State Management: Zustand
- Animations: Framer Motion
- Backend: FastAPI (Python)
- Database: Supabase (PostgreSQL)
- Payments: Flutterwave integration
- Remote Desktop: noVNC over WebSocket
- VM: Docker containerization

ACTION TYPES REFERENCE

Control supports a comprehensive set of action types for automation:

Action Type	Description	Parameters
click	Single left-click at coordinates	x, y (0-1000)
double_click	Double-click at coordinates	x, y (0-1000)
right_click	Right-click context menu	x, y (0-1000)
mouse_move	Move cursor to position	x, y (0-1000)
drag	Click and drag operation	start, end coordinates
scroll	Scroll at position	x, y, direction, amount

type	Type text string	text, speed (optional)
key_press	Press key combination	key (e.g., 'ctrl+c')
terminal	Execute shell command	command string
browser_open	Open URL in browser	url
browser_screenshot	Capture browser content	none
browser_execute_js	Run JavaScript	code string
wait	Pause execution	duration (seconds)
screenshot	Capture screen region	optional region

CONCLUSION

Control represents a significant advancement in desktop automation technology. By combining computer vision, advanced AI reasoning, and comprehensive input control, it enables users to automate virtually any desktop task through natural language commands. The dual-component architecture provides both local automation capabilities and cloud-based VM management, addressing a wide range of use cases from personal productivity to enterprise automation.

The platform's commitment to a local-first approach, multi-provider AI support, and robust security measures positions it as a compelling solution for users seeking powerful automation without compromising privacy or flexibility.